Tag-Constrained Pseudo-Labels and Two-Decode Voting for Noise Event Detection in Indic Speech

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***Abstract*—We detect the onset and offset of every non-speech noise event in Vaani speech clips (Combined = event F1 + segment Dice), with scarce verified labels and a test set that mixes natural recordings with synthetic mixtures. Two systems share three components. First, the event tags in Vaani transcripts fix how many events a teacher ensemble may label in a clip, and roughly where, which improves pseudo-label quality on a 305k-clip transcribed pool (+0.050 against Gold). Second, BEATs and Dasheng encoders learn event boundaries through a localisation pretext task on unlabelled Vaani audio. Third, every model is decoded twice, by a threshold with a learned boundary refiner and by an event-proposal head, and a weighted vote fuses all decodes. Our router-free system runs five students trained on blended natural and synthetic data on every clip and scores 1.6114; nothing in it reads a clip's domain, so one model set and one vote serve natural recordings and mixtures alike. Our routed system detects the organisers' mixtures by their peak level and sends them to synthetic specialists; it placed second with 1.6168, 0.0004 behind first. Both reach event F1 0.53 on natural clips and 0.99 on mixtures. Code: https://github.com/mcjoshi/indic-noise-event-detection; checkpoints: https://huggingface.co/manishjoshiai/indic-noise-event-detection.**

I. Training Data and Preprocessing

Task and metric. Each clip may contain several non-speech noise events. Event F1 counts a prediction as correct when both its onset and offset lie within $\max(0.2\,d, 0.05)$ s of a reference event of duration d [1]; segment Dice compares 10 ms frames; Combined is their sum (maximum 2).

Competition data. The release (ARTPARK-IISc/Vaani-Noise-Event-Dataset) contains 90,635 labelled clips (154.6 h): 11,111 Gold clips with verified timestamps, 61,640 Silver and 17,884 Bronze. We hold out a speaker-disjoint split of 2,005 Gold clips (4.0 h) for selection, removing 1,163 further clips of the same speakers from training; three late models also train on it (Sec. III). The organisers' validation release adds 1,806 natural and 695 synthetic clips, which several models also train on (Sec. III). Audio is 16 kHz mono; models see random crops of up to 15 s as 128-bin log-mel features on a 10 ms grid.

External data. (i) 321,728 transcribed clips of ARTPARK-IISc/Vaani-transcription-part [2] (539 h), 305,190 of which received tag-constrained labels, excluding every competition clip and every recording that contains a validation or test clip; (ii) 207,268 untranscribed clips of ARTPARK-IISc/Vaani (304 h), excluding competition clips by clip ID, used for the pretext task and early pseudo-labels; (iii) ESC-50 [3] (ashraq/esc50 mirror, CC BY-NC) events for mixtures and the pretext task; (iv) the Track 2 noisy/clean training pairs, only to train the enhancement U-Net (Table I). No test labels, test pseudo-labels or manual annotation were used.

Tag-constrained pseudo-labels. Vaani transcribers mark audible events with paired tags such as `<horn>...</horn>`; on our held-out Gold split the number of event tags equals the number of verified events on 95.8% of clips. A teacher ensemble proposes candidate spans with vote counts (IoU 0.6 clustering, median edges). We discard non-event tags (`<noise>`, `<PAUSE>`, `[unintelligible]`), map each remaining tag's relative character position to a time prior, and assign exactly one candidate per tag by Hungarian matching on vote support and distance to the prior (Fig. 2a); unmatched tags are left empty. With the first-round teacher (three models, six decodes) this raises label quality against Gold from 1.1903 to 1.2403 Combined; the five-model teacher used for the final students reaches 1.2578. The labels cover the transcribed clips and the Bronze tier; Silver edges move onto a fused span when the two overlap at IoU $\geq$0.3.

Synthetic mixtures. We insert up to six events, non-overlapping except in our earliest ESC-50 set, into the event-free stretches of Gold training clips, kept at natural level and peak-normalised afterwards. Events come from ESC-50 folds 1–4 and Vaani Gold events (168k clips, 6–12 dB), or from 2,213 noise events obtained by subtracting the clean references supplied with the organisers' synthetic validation clips (48k clips, -6 to 13 dB). Router-free students train on roughly balanced blends of natural and synthetic clips (147k, 200k or 240k rows; the 200k blend adds our validation split and the organisers' natural clips). Augmentation: random crops, threefold Gold oversampling for natural-route models, mixup ($\alpha = 0.2$, max-union targets) for the BEATs and Dasheng specialists, and shift, gain and noise views for a consistency loss.

II. Proposed Method

Architecture. Every model is a CRNN [4] on the 10 ms grid: seven Conv–BN–GELU blocks pooled over frequency only, concatenated with projected encoder frames (LayerNorm, linear to 256, interpolated to 10 ms), then a two-layer BiGRU. Whisper and WavLM features are a learned softmax-weighted sum of all layers [5]; BEATs and Dasheng use their last layer.

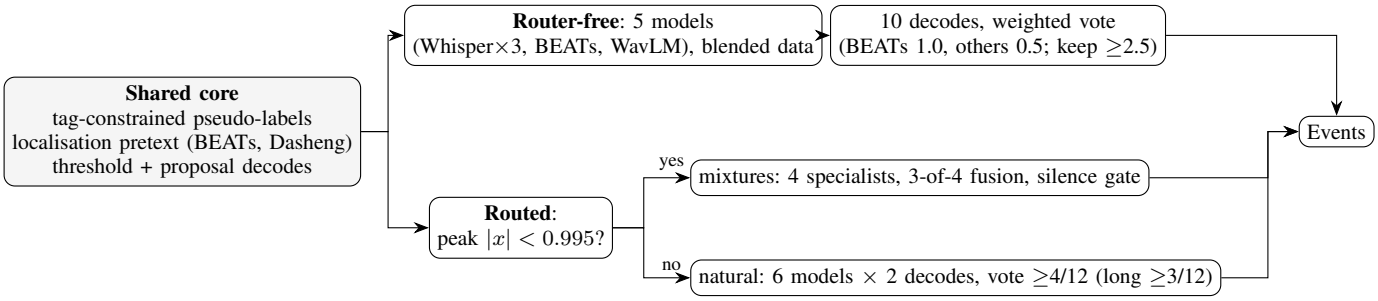


Fig. 1: Both systems are built on the same core. The router-free system (top) runs one model set and one vote on every clip; the routed system (bottom) separates the organisers' mixtures by peak level and serves each domain with its own models.

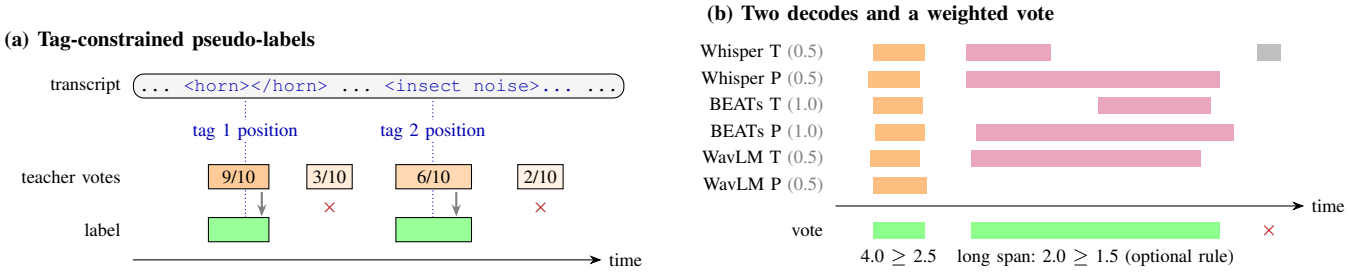


Fig. 2: The shared core, schematically. (a) Tag count fixes how many events are kept; each tag's position in the text gives a time prior, and Hungarian matching keeps one teacher candidate per tag. (b) T = threshold decode with refiner, P = proposal decode (two of the three Whisper students omitted); a span survives at summed weight 2.5, long spans at 1.5 under the optional rule, and an isolated span (0.5) is dropped.

Heads predict frame activity, onset/offset bumps, a second overlap layer, seven auxiliary categories, a 100 ms occupancy branch, and an Event Proposal Network (EPN) [6] in which every active frame regresses its distance to the onset and offset of its event.

Losses and training. The objective sums frame BCE on soft coverage targets (non-Gold background down-weighted), onset/offset BCE, a clip-level multiple-instance BCE (the only signal taken from Bronze), category BCE and a duration-normalised $1 - \mathrm{IoU}$ EPN loss (weight 2). Most models add a Gold term on frames inside events of at least 4 s (weight 1, or 3 in two models), and some add soft Dice or mean-teacher consistency [7]. Gold carries weight 1, Silver 0.5–0.9 and pseudo-labels 0.35–0.7. We use AdamW (weight decay 10^{-2}), warmup and cosine decay, bf16, gradient clipping 5 and EMA 0.999, and select checkpoints on validation Combined. Encoders are fine-tuned at a tenth of the head rate (0.85 layer-wise decay for WavLM, BEATs and Dasheng). Training proceeds in chains in the spirit of noisy-student self-training [8] and multi-stage ensemble distillation for SED [9]: a frozen-encoder parent, an unfrozen child, then a one-epoch student on teacher-labelled data; the 15 final models descend from 14 ancestors, all released with their exact configurations.

Localisation pretext. BEATs and Dasheng first spend 10k steps localising 0–3 ESC-50 events pasted into 8 s crops of untranscribed Vaani audio; the boundary hit rate on held-out inserts rises from 0.059 to 0.706 (BEATs) and 0.728 (Dasheng).

Two decodes per model. The threshold decode applies a tuned threshold, median filter, merge gap, minimum duration and edge offsets; a small Conv1d refiner then predicts a 101-bin distribution over ±250 ms around each endpoint from the model's own curves, and the endpoint moves to its expected value. The proposal decode selects EPN intervals greedily by presence and suppresses overlaps; the two decodes fail differently (Fig. 2b).

Router-free system. Five students (three Whisper-medium, one BEATs-pretext, one WavLM) run on every clip. Their ten decodes are clustered at IoU 0.6 with median edges; Whisper and WavLM decodes weigh 0.5 and BEATs decodes 1.0 (chosen on validation), and a cluster survives at summed weight 2.5 of 6. Nothing in this system reads a clip's domain: one model set, one decode configuration per model and one vote serve natural recordings and mixtures alike. An optional long-span rule keeps clusters of at least 6 s at weight 1.5 (IoU 0.45); it improved F1 and Dice on both halves of our validation split (measured on the previous model set) but has no board score.

Routed system. The organisers' mixtures are peak-normalised to 0.99 while natural clips reach full scale: on the test set 2,080 clips peak at 0.99 and 3,437 at 1.0, exactly the stated subset sizes, a property of the release that a peak test ($|x| < 0.995$) exploits. Mixtures go to four specialists (BEATs with an enhancement-residual input channel, two Dasheng seeds, Whisper) fused at three of four votes [10], followed by a gate that drops spans more than 30 dB below the clip's 95th-percentile level. Natural clips are fused from six models × two decodes, keeping clusters with four of twelve votes, or three when the span is at least 4 s (IoU 0.45).

TABLE I: Pretrained models (Hugging Face IDs unless noted) and the number of routed (R) and router-free (U) models using each.

Model	Version / source	Role
Whisper-medium [11]	openai/whisper-medium	4 R, 3 U
WavLM-base-plus [12]	microsoft/wavlm-base-plus	1 R, 1 U
HuBERT-large [13]	facebook/hubert-large-ll60k	WavLM ancestor
BEATs iter3+ AS2M [14]	BEATs_iter3_plus_AS2M.pt, microsoft/unilm	3 R, 1 U; pretext
Dasheng base [15]	pip dasheng 0.0.9 (Xiaomi)	2 R; pretext
Enhancement U-Net	ours (Track 2 pairs), released	1 R, residual

TABLE II: Phase 2 test set, event F1 / Dice overall and per subset; – = not shown. †Derived: Dice as the clip-weighted mean of subset Dice (exact for our routed entry), F1 = Combined − Dice.

System	Combined	F1 / Dice	Natural	Synthetic
1st place	1.617289	.789 / .828	–	–
Ours, routed (2nd)	1.616847	.790 / .827	.526 / .726	.997 / .992
3rd place	1.616833	.789 / .828	–	–
Ours, router-free	1.6114	.793 / .819†	.526 / .715	.993 / .990

III. Results and Conclusion

Leaderboard. In Table II the routed system is second, 0.0004 below first and 0.000014 above third; both gaps lie inside the ±0.002 spread between seeds of one model slot, so the order of the top three is not meaningful. The router-free system is 0.0054 lower (fourth as a separate entry; fourth place 1.611179), yet its derived event F1 (0.793) is above each of the top three and its natural F1 equals the routed system's: the gap is Dice (0.819 vs 0.827), mostly on natural clips, where the systems also differ in models and vote rule.

Component evidence. In Table III the tag gain comes from position matching, not counts alone; for a single student, tag labels add 0.0036 and the extra transcribed audio 0.0052.

Disclosures. Late changes to both systems were selected on the Phase 2 board within its daily limit, so the board deltas in Table III and Fig. 3 are optimistic. The organisers' 1,806 natural clips train one BEATs and two Whisper natural models (routed) and one Whisper student (router-free); the last three also train on our validation split and select their checkpoints on it. All 695 synthetic clips train every synthetic specialist and every router-free student, so scores on those clips are in-sample. Unlabelled test audio was used twice: its acoustic statistics guided the background augmentation of our earliest mixture set, and its peak values checked the router. The untranscribed pool (ii) was filtered by clip ID only; 108 of its 207,268 clips come from recordings that also contain a test clip, without labels. From raw audio, the public code reproduces both router-free submissions on all 5,517 test clips and the routed one on 5,516 (one submitted clip carried a span patched in from an earlier submission).

Conclusion. Transcript tags are a cheap way to constrain pseudo-labels for noise event detection, and with a localisation pretext and two complementary decodes per model they support both a routed system that placed second and a router-free system 0.0054 behind it. Natural recordings remain the open problem (event F1 0.53 against 0.99 on mixtures).

TABLE III: Component evidence. L = pseudo-label quality vs Gold, V = our validation, O = organisers' natural clips, B = Phase 2 board; Single = the only change between the two runs.

Change (base)	Metric	Single	Δ
Shared core			
Tag count + position matching (teacher)	L	yes	+0.050
Tag count only, top-N (teacher)	L	yes	+0.003
Student on tag vs plain labels (1 model)	V	yes	+0.0036
Transcribed Vaani audio added (1 model)	V	yes	+0.0052
No proposal decode, 4 of 6 teachers (6-model teacher)	L	yes	−0.014
Localisation pretext (1 BEATs model)	V / O	yes	−0.003 / +0.017
Router-free system			
30 dB gate on every clip	V	yes	−0.017
Long-span rule (both halves)	V	yes	+0.003 Dice
Routed system			
Whisper synthetic specialist	B	yes	+0.0006
Long clusters: 3 votes, IoU 0.45	B	2 steps	+0.0005
Routed vs router-free (models differ)	B	no	+0.0054

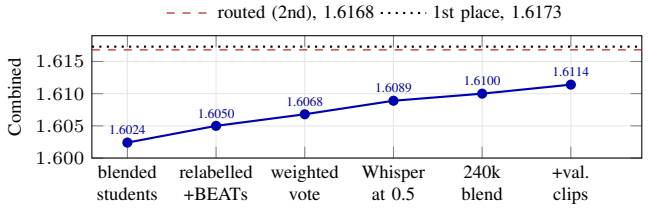


Fig. 3: Router-free system on the Phase 2 board, one submission per point. Steps 2, 3 and 5 change several factors at once (step 3 also relabels the BEATs student; step 5 also re-weights one Whisper student).

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